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Confidently Wrong: Silent Memory Failures in LLMs Under Interference



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Kanak Raj (/profile?id=~Kanak_Raj1),
Sourav Chattaraj (/profile?id=~Sourav_Chattaraj1)

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Keywords: Large Language Models, Memory Interference, Epistemic Intelligence, Proactive Interference, Retroactive Interference, Silent Failure, Calibration, In-Context Retrieval
TL;DR: LLMs fail silently under in-context interference: 97% of wrong retrievals across 39 models produce no uncertainty signal, with a structured RI/PI asymmetry in metacognition.

Abstract:
Across 39 large language models, we identify a silent memory failure under in-context interference: models confidently return wrong values with fewer than 1% hallucination. The silence has structure: RI errors are fully silent (0% refusal), while PI errors carry a weak uncertainty signal (~4% refusal) — an asymmetry in when models flag a failure. We adapt the classical AB-AC paired-associate interference paradigm (Barnes & Underwood, 1959) — 46 category-value pairs followed by N interleaved updates per category (N in {3, 10, 50, 100, 200, 300}) — to test retrieval of either the initial value (retroactive interference, RI) or the most recent value (proactive interference, PI) on identical stimulus sequences. Every model exhibits the same pattern: PI causes substantially more degradation than RI (Cohen's d = 1.73, p < 0.0001), inverting the typical human pattern. The failure has additional structure: model size predicts RI resistance ($R^2 = 0.49$) but not PI resistance ($R^2 = 0.06$, n.s.), and chain-of-thought reasoning models improve RI while worsening PI. Error analysis reveals distinct failure modes: RI failures are passive retrieval failures (51%), while PI failures show active primacy intrusion (56%). These results characterize a structured epistemic blind spot: LLMs have partial self-knowledge about their memory failures, asymmetric between the two interference types, with implications for reliability in interference-heavy deployment settings.

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Decision by Program Chairs 19 May 2026, 13:58 Program Chairs, Area Chairs, Reviewers, Authors

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Review of paper 55 Confidently Wrong: Silent Memory Failures in LLMs Under Interference

Official Review by Reviewer kL7u  15 May 2026, 13:24 (modified: 18 May 2026, 17:39)

 Program Chairs, Area Chairs, Reviewers, Authors  Revisions (/revisions?id=1daOibA6wq)

Summary:

This paper asks whether LLMs fail silently when asked to retrieve a specific value from a long sequence of conflicting updates. The experimental assessment shows that they do, consistently, across models of varying scale and architecture. The experimental design is borrowed from a paired-associate learning paradigm from cognitive psychology, aiming to check models on retroactive/proactive interference. The paper shows that retrieving the most recent value in the category-value pairs degrades far more severely than retrieving the original, inverting the human recency-dominance pattern. While model size helps with retroactive interference, and chain-of-thought reasoning models improve it further, both factors offer no advantage (and in the case of reasoning models, actively worsen) proactive interference.. Interestingly, models almost never flag their errors. The authors also show that the two types of interference fail in qualitatively different ways, the former through passive inaccessibility of the target value, the latter through active intrusion of earlier ones.

Strengths:

The empirical setting is appropriate, with experiments on 39 models across seven families, spanning several orders of magnitude in parameter count. Also the design is well chosen: borrowing the AB-AC paradigm provides theoretical grounding, and using identical stimulus sequences for both conditions cleanly isolates interference from positional search difficulty. The more substantive contribution of the paper is the structural characterisation of the failure. The three dissociations (by model size, by reasoning architecture, and by metacognitive signal) argue against a unified capacity account and point toward distinct underlying processes. The error taxonomy adds further resolution: passive retrieval failure for RI versus active primacy intrusion for PI, with near-zero hallucination in both conditions. This reframes interference as a retrieval and selection problem rather than a generation one, which is both conceptually precise and practically useful.

Weaknesses:

All results derive from a single AB-AC category-value task with artificial stimuli, and it is unclear how far the RI/PI dissociation generalises to naturalistic settings where updates are non-uniform, semantic similarity varies, and the model carries prior world-knowledge about the categories. The mechanistic discussion seems to outpace the evidence. The superposition hypothesis and primacy protection account are framed as explanations rather than post-hoc hypotheses, and the hedging language is easy to miss. A single attention-probing experiment on an open-weight model would have been more persuasive than the current narrative. The 0% PI result for reasoning models is a practically significant finding, supported by one illustrative example. A more systematic analysis, covering error position distributions, failure onset across interference levels, and behaviour at temperature above zero, is needed before a claim of this magnitude can be considered well-supported. The hallucination definition is narrower than is standard in the literature, classifying earlier-value intrusions as positional errors rather than hallucinations. This choice is internally consistent but is not entirely consistent with the typical definition.

Limitations:

The ethical considerations section acknowledges dual-use concerns but only briefly. The paper provides a ranked list of model vulnerabilities with precise interference thresholds at which different models fail. For practitioners deploying models in the high-stakes settings the introduction invokes, specifically medical records and legal documents, this information is directly actionable so this might warrant a more substantive treatment.

Rating: 5: Borderline accept

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Straightforward Study

Official Review by Reviewer iddL  28 Apr 2026, 14:58 (modified: 18 May 2026, 17:39)

 Program Chairs, Area Chairs, Reviewers, Authors  Revisions (/revisions?id=dOhaH1cw78)

Summary:

The paper [Confidently Wrong: Silent Memory Failures in LLMs Under Interference] provides an empirical study of in-context value retrieval failures in LLMs. In particular, the author suggest a study design in which an LLM is prompted with category-value pairs, modifications to those pairs and then the retrieval question for the first or last value of the category-value pair. Different to human recall studies which find a recency bias, LLMs are better in re-producing initial values of the prompt.

Strengths:

The paper makes the interesting observation that the recall pattern for LLMs is inverted in comparison to humans. I didn't find any substantial error.

Weaknesses:

The paper is rather technically written, but understandable. I list some minor aspects which could improve the presentation: a) Figure 2 has overlapping texts and graphical elements. b) The references don't seem to be fully checked. E.g., Barnes and Underwood, 1959, is entitled " 'Fate' of first-list associations in transfer theory" and not "Fate of first-list associations in transfer theory". c) The authors don't define Proactive Interference and Retroactive Interference properly. It is written: "recalling the initial value of a key after N interleaved updates (retroactive interference, RI) or recalling the latest value (proactive interference, PI)." However, in this sentence the task is described and not "interference" explained. This becomes confusing in the caption of Figure 1 where $PI > RI$. Because here PI and RI are negatively connotated terms somewhat inversely related to the depicted accuracy. Note furthermore that in the psychological literature the authors refer to, e.g., [Barnes and Underwood, 1959. "Fate" of first-list associations in transfer theory] or [Murdock, 1962. The serial position effect of free recall], the "I" stands for "inhibition" and not "interference" (cf. [Young, 1955. Retroactive and proactive effects under varying conditions of response similarity]). d) It would be nice to see an explanation of the fact that LLama-3.2-3b performed 0% accuracy at N=3, (Table 1), which seems surprising. e) I don't really understand why Figure 1 seems to average over all 39 models. This indeed gives a nice picture, but it does not seem meaningful to me. I'm not interested in RI or PI on average over all models, as I'm always only using a single model. I think it would be better to show the curve of a single model here. f) How about the order of the initial learning? Is this fixed or as well is it equipped with a randomized order?

In terms of the content of the paper I have to admit I'm not an expert on LLMs. So for instance, I'm not familiar with existing literature on in-context retrieval success of LLMs. The experimental setup is rather simple. There are several open questions not addressed however: a) What is the dependency of the retrieval success on the instruction? (cf. [Sclar et al., 2023. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting]) b) What about retrieval success for middle-position values? (cf. the RI-curve as shown in [Murdock, 1962. The serial position effect of free recall]) c) What are the exact statistical effects of the bootstrapping methodology? Are there comparable studies who used a similar trick? That methodology seemed fishy on a first glance, but might well be justified. I think I didn't really understand the justification.

Limitations:

The limitations address the main concern I have: external validity of the results. In which ways is the insight of the paper relevant and valid in non-artificial retrieval settings. I think if there is a concrete example in which the retrieval setting is well-motivated it could be used to strengthen the introduction and motivate the study.

Rating: 5: Borderline accept

Confidence: 2: You are willing to defend your assessment, but it is quite likely that you didnot understand the central parts of the submission or that you are unfamiliar with some pieces of related work.

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